



Cross-Border IPS AI Agent

Federated terminology alignment + FHIR-native trust + validated IPS-style delivery

TECH ADAPTIVE PANDIT

Anant Gupta | Manish Gupta | Gourav Gupta

Efficiency & Effectiveness | Synthetic Type 2 diabetes case study

4 sites

India | USA | Australia | Europe

768

local terminology mentions

48/48

federated vs 47/48 local-only

192/192

globally unseen | macro-F1 1.000

4/4 | 0 errors

IPS 2.0.1 validator | 0 warnings

1 | LOCAL CLINICAL SITE

FL training records stay at the site

INPUT MODE

Structured EHR -> direct facts
Free text -> rule-backed extraction

CLINICAL FACT MODEL

Source span retained with
condition, lab, value, medication,
dose, and context

REGISTRY HIT

Trusted code

REGISTRY MISS

Federated linker

Confidence <0.70: human review; no clinical code emitted

2 | SEMANTIC INTEROPERABILITY

FHIR terminology makes mapping auditable

LOCAL CODESYSTEM + VALUESET

Represent source phrase and
constrain target concepts by
clinical category

CONCEPTMAP + \$TRANSLATE

Publish source-to-target mapping,
equivalence, provenance, and
confidence

\$LOOKUP + \$VALIDATE-CODE

Check target-code existence and
allowed ValueSet membership

Code-valid does not mean semantically correct: gold labels + clinician review

3 | DATA INTEROPERABILITY

FHIR IPS is the exchange artifact

FHIR IPS DOCUMENT BUNDLE

Bundle.type=document | Composition
first | persistent identifier

CODED FHIR RESOURCES

Patient | Condition | Observation |
MedicationStatement | Organization

RECEIVER READINESS

US Core STU9 | ABDM R4 draft | AU
Core CI | EPS ballot

PDF = human rendering; FHIR IPS = interoperable artifact

GENUINE FEDERATED TERMINOLOGY LEARNING

Each simulated site trains locally; the coordinator never receives reports or patient-level Bundles.

INDIA

local SGD

USA

local SGD

AUSTRALIA

local SGD

EUROPE

local SGD

FEDAVG COORDINATOR

weighted parameters

sample-count weighting

Round loop: local SGD -> weight update -> FedAvg -> global model broadcast

Upload payload: model tensors + sample count only

48.05 KiB/client update | 1.88 MiB two-way traffic across 5 rounds

Data locality, not cryptography: updates may leak without secure aggregation or differential privacy.

ONE TRACE, TWO VALIDATION LAYERS

Semantic mapping is tested separately from FHIR structure/profile validation.

SYNTHETIC SOURCE

madhumeha type 2

CONCEPT

Type 2 diabetes mellitus

CODES

SNOMED 44054006
ICD-10 E11

FHIR

Condition

Semantic evidence: 48/48 transfer at all 5 seeds; 192/192 globally unseen; macro-F1 1.000.

FHIR evidence: 4/4 Bundles pass Validator 6.9.11 against IPS 2.0.1 with 0 errors and 0 warnings.

Terminology snapshot: representative SNOMED CT, LOINC, RxNorm, and ICD-10 codes pass 4/4 checks.

CLAIM BOUNDARY Proven here: synthetic FL benchmark, FHIR-native artifacts, official validator evidence, and downloadable route outputs.

Not claimed: clinical accuracy, national certification, formal privacy, production readiness, or completed independent clinician sign-off.